Statistical Foundations of Machine Learning

Fast Paced TA Session – Problems & Solutions

6.30 – 7.30 pm, 24th Oct 2026

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1. Normal Random Variables

Problem 1.1 - Covered in previous TA session

Problem:
If X is a normal random variable with mean $\mu = 3$ and variance $\sigma^2 = 16$, find:

(a) $P\{X < 11\}$

(b) $P\{X > -1\}$

(c) $P\{2 < X < 7\}$

Solution:
Here, $\mu = 3$ and standard deviation $\sigma = \sqrt{16} = 4$. We standardize using $Z = \dfrac{X-3}{4} \sim \mathcal{N}(0,1)$.

(a) Standardizing $X = 11$:

$$P(X < 11) = P\left(Z < \frac{11-3}{4}\right) = P(Z < 2) = \Phi(2.00) \approx \mathbf{0.9772}$$

(b) Standardizing $X = -1$:

$$P(X > -1) = P\left(Z > \frac{-1-3}{4}\right) = P(Z > -1) = P(Z < 1) = \Phi(1.00) \approx \mathbf{0.8413}$$

(c) Standardizing the lower and upper bounds:

$$z_1 = \frac{2-3}{4} = -0.25, \qquad z_2 = \frac{7-3}{4} = 1.00$$

$$P(2 < X < 7) = \Phi(1.00) - \Phi(-0.25) = \Phi(1.00) - [1 - \Phi(0.25)]$$

From normal tables, $\Phi(1.00) \approx 0.8413$ and $\Phi(0.25) \approx 0.5987$:

$$P(2 < X < 7) \approx 0.8413 - (1 - 0.5987) = 0.8413 - 0.4013 = \mathbf{0.4400}$$

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Problem 1.2

Problem:
Data from the National Oceanic and Atmospheric Administration indicate that the yearly precipitation in a certain coastal region is normally distributed with a mean of 12 inches and a standard deviation of 3 inches.

(a) Find the probability that the total precipitation during the next 2 years will exceed 25 inches.

(b) Find the probability that next year's precipitation will exceed that of the following year by more than 3 inches.

Assume that the precipitation totals for the next 2 years are independent.

Solution:
Let X_1 and X_2 denote the precipitation in year 1 and year 2, respectively, where $X_1, X_2 \overset{\text{iid}}{\sim} \mathcal{N}(\mu = 12, \sigma^2 = 9)$.

(a) Let $T = X_1 + X_2$ be the total precipitation. Since the sum of independent normal random variables is normally distributed:

$$E[T] = E[X_1] + E[X_2] = 12 + 12 = 24 \text{ inches}$$

$$\text{Var}(T) = \text{Var}(X_1) + \text{Var}(X_2) = 9 + 9 = 18 \implies \sigma_T = \sqrt{18} \approx 4.243$$

Standardizing for $T > 25$:

$$P(T > 25) = P\left(Z > \frac{25 - 24}{\sqrt{18}}\right) = P\left(Z > \frac{1}{4.243}\right) \approx P(Z > 0.24)$$

From standard normal tables:

$$P(T > 25) = 1 - \Phi(0.24) \approx 1 - 0.5948 = \mathbf{0.4052}$$

(b) Let $D = X_1 - X_2$ be the difference between the two years:

$$E[D] = E[X_1] - E[X_2] = 12 - 12 = 0$$

$$\text{Var}(D) = \text{Var}(X_1) + (-1)^2\text{Var}(X_2) = 9 + 9 = 18 \implies \sigma_D = \sqrt{18} = 3\sqrt{2}$$

Standardizing for $D > 3$:

$$P(D > 3) = P\left(Z > \frac{3 - 0}{\sqrt{18}}\right) = P\left(Z > \frac{1}{\sqrt{2}}\right) \approx P(Z > 0.71)$$

From standard normal tables:

$$P(D > 3) = 1 - \Phi(0.71) \approx 1 - 0.7611 = \mathbf{0.2389}$$

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2. Sample Mean and Central Limit Theorem

Problem 2.1

Problem:
Approximate the probability that the sum of 16 independent uniform $(0, 1)$ random variables exceeds 10.

Solution:
Let $X_1, X_2, \ldots, X_{16} \overset{\text{iid}}{\sim} \text{Uniform}(0, 1)$. For each X_i:

$$E[X_i] = \frac{0+1}{2} = 0.5, \qquad \text{Var}(X_i) = \frac{(1-0)^2}{12} = \frac{1}{12}$$

Let $S_{16} = \sum_{i=1}^{16} X_i$ be their sum:

$$E[S_{16}] = 16 \times 0.5 = 8$$

$$\text{Var}(S_{16}) = 16 \times \frac{1}{12} = \frac{4}{3} \implies \sigma_{S_{16}} = \sqrt{\frac{4}{3}} = \frac{2}{\sqrt{3}} \approx 1.1547$$

By the Central Limit Theorem, $S_{16} \approx \mathcal{N}\left(8, \frac{4}{3}\right)$. Standardizing:

$$P(S_{16} > 10) = P\left(Z > \frac{10-8}{2/\sqrt{3}}\right) = P(Z > \sqrt{3}) \approx P(Z > 1.732)$$

From standard normal tables:

$$P(S_{16} > 10) = 1 - \Phi(1.732) \approx 1 - 0.9584 = \mathbf{0.0416}$$

Problem 2.2

Problem:
The amount of time that a certain type of battery functions is a random variable with mean 5 weeks and standard deviation 1.5 weeks. Upon failure, it is immediately replaced by a new battery. Approximate the probability that 13 or more batteries will be needed in a year (take 1 year = 52 weeks).

Solution:
Let X_i be the lifespan in weeks of the i-th battery, with $\mu = 5$ and $\sigma = 1.5$.

At least 13 batteries will be needed in 52 weeks if and only if the first 12 batteries fail before the end of the year. That is, $S_{12} = \sum_{i=1}^{12} X_i < 52$.

$$E[S_{12}] = 12 \times 5 = 60 \text{ weeks}$$

$$\text{Var}(S_{12}) = 12 \times (1.5)^2 = 12 \times 2.25 = 27 \implies \sigma_{S_{12}} = \sqrt{27} \approx 5.196$$

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By the Central Limit Theorem, $S_{12} \approx \mathcal{N}(60, 27)$. Standardizing:

$$P(S_{12} < 52) = P\left(Z < \frac{52 - 60}{\sqrt{27}}\right) = P\left(Z < \frac{-8}{5.196}\right) \approx P(Z < -1.54)$$

From standard normal tables:

$$P(S_{12} < 52) = 1 - \Phi(1.54) \approx 1 - 0.9382 = \mathbf{0.0618}$$

Problem 2.3

Problem:
The lifetime of a certain electrical part is a random variable with mean 100 hours and standard deviation 20 hours. If 16 such parts are tested, find the probability that the sample mean is:

(a) less than 104 hours;

(b) between 98 and 104 hours.

Solution:
Let $\bar{X}$ be the sample mean of $n = 16$ independent parts.

$$E[\bar{X}] = \mu = 100, \qquad \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{20}{\sqrt{16}} = \frac{20}{4} = 5 \text{ hours}$$

By the Central Limit Theorem, $\bar{X} \approx \mathcal{N}(100, 5^2)$.

(a) Standardizing $\bar{X} = 104$:

$$P(\bar{X} < 104) = P\left(Z < \frac{104 - 100}{5}\right) = P(Z < 0.80) = \Phi(0.80) \approx \mathbf{0.7881}$$

(b) Standardizing both bounds:

$$P(98 \le \bar{X} \le 104) = P\left(\frac{98 - 100}{5} \le Z \le \frac{104 - 100}{5}\right) = P(-0.40 \le Z \le 0.80)$$

$$= \Phi(0.80) - \Phi(-0.40) = \Phi(0.80) - [1 - \Phi(0.40)]$$

From normal tables, $\Phi(0.80) \approx 0.7881$ and $\Phi(0.40) \approx 0.6554$:

$$P(98 \le \bar{X} \le 104) \approx 0.7881 - (1 - 0.6554) = 0.7881 - 0.3446 = \mathbf{0.4435}$$

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3. Maximum Likelihood Estimation (MLE)

Problem 3.1

Problem:
Let $X_1, X_2, \ldots, X_n$ be an independent and identically distributed (i.i.d.) random sample from an exponential distribution with unknown rate parameter $\lambda > 0$, whose probability density function is given by:

$$f(x;\lambda) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Determine the maximum likelihood estimator (MLE) of λ.

Solution:

The likelihood function $L(\lambda)$ for the sample $X_1, \ldots, X_n$ is:

$$L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda X_i} = \lambda^n \exp\left(-\lambda \sum_{i=1}^{n} X_i\right)$$

Taking the natural logarithm yields the log-likelihood:

$$\ln L(\lambda) = n \ln \lambda - \lambda \sum_{i=1}^{n} X_i$$

Differentiating with respect to λ and setting the derivative to zero:

$$\frac{d}{d\lambda} \ln L(\lambda) = \frac{n}{\lambda} - \sum_{i=1}^{n} X_i = 0 \implies \hat{\lambda}_{\text{MLE}} = \frac{\mathbf{n}}{\sum_{\mathbf{i=1}}^{\mathbf{n}} \mathbf{X_i}} = \frac{\mathbf{1}}{\bar{\mathbf{X}}}$$

To verify that this point is a maximum, check the second derivative:

$$\frac{d^2}{d\lambda^2} \ln L(\lambda) = -\frac{n}{\lambda^2} < 0 \quad \text{for all } \lambda > 0$$

Thus, $\hat{\lambda}_{\text{MLE}} = \frac{1}{\bar{X}}$ uniquely maximizes the likelihood.

Problem 3.2

Problem:
Let $X_1, X_2, \ldots, X_n$ be an i.i.d. random sample from a Laplace distribution with unknown location parameter $\mu \in \mathbb{R}$ and known scale parameter $b = 1$, whose probability density function is given by:

$$f(x;\mu) = \frac{1}{2} e^{-|x-\mu|}, \quad x \in \mathbb{R}$$

Determine the maximum likelihood estimator of the location parameter μ.

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Solution:
The joint likelihood function $L(\mu)$ for the observations $X_1, \ldots, X_n$ is:

$$L(\mu) = \prod_{i=1}^{n} \frac{1}{2} e^{-|X_i - \mu|} = \left(\frac{1}{2}\right)^n \exp\left(-\sum_{i=1}^{n} |X_i - \mu|\right)$$

Taking the log-likelihood:

$$\ln L(\mu) = -n \ln 2 - \sum_{i=1}^{n} |X_i - \mu|$$

Because $-n \ln 2$ is a constant, maximizing $\ln L(\mu)$ is mathematically equivalent to minimizing the sum of absolute deviations (L_1 loss):

$$S(\mu) = \sum_{i=1}^{n} |X_i - \mu|$$

Let the ordered sample values be denoted by $X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}$.

Evaluating the derivative (where defined) via the sign function:

$$\frac{d}{d\mu} S(\mu) = -\sum_{i=1}^{n} \operatorname{sgn}(X_i - \mu) = (\text{number of observations } < \mu) - (\text{number of observations } > \mu)$$

The derivative changes sign when the number of observations on either side of μ is equalized:

- **When n is odd** ($n = 2m+1$): The minimum is uniquely attained at the middle observation:

$$\hat{\mu}_{\text{MLE}} = \mathbf{X_{(m+1)}} = \text{Sample Median}(\mathbf{X_1}, \ldots, \mathbf{X_n})$$

- **When n is even** ($n = 2m$): The derivative is zero throughout the closed interval between the two central order statistics $[X_{(m)}, X_{(m+1)}]$. Any value in this interval minimizes $S(\mu)$:

$$\hat{\mu}_{\text{MLE}} \in [\mathbf{X_{(m)}}, \mathbf{X_{(m+1)}}]$$

  By standard convention, the midpoint is taken: $\hat{\mu}_{\text{MLE}} = \dfrac{X_{(m)} + X_{(m+1)}}{2}$.

Thus, the MLE of the Laplace location parameter is the **sample median**.

Problem 3.3

Problem:
Let $X_1, \ldots, X_n$ be a sample from a normal μ, σ^2 population. Determine the maximum likelihood estimator of σ^2 when μ is known. What is the expected value of this estimator?

Solution:
Let $\theta = \sigma^2$. Since μ is known, the likelihood function is:

$$L(\theta) = (2\pi\theta)^{-n/2} \exp\left(-\frac{1}{2\theta} \sum_{i=1}^{n} (X_i - \mu)^2\right)$$

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Taking the log-likelihood:

$$\ln L(\theta) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln\theta - \frac{1}{2\theta}\sum_{i=1}^{n}(X_i - \mu)^2$$

Differentiating with respect to θ and setting to zero:

$$\frac{d}{d\theta}\ln L(\theta) = -\frac{n}{2\theta} + \frac{1}{2\theta^2}\sum_{i=1}^{n}(X_i - \mu)^2 = 0$$

$$\implies \hat{\sigma}^{\mathbf{2}}_{\mathrm{MLE}} = \frac{\mathbf{1}}{\mathbf{n}}\sum_{\mathbf{i=1}}^{\mathbf{n}}(\mathbf{X_i} - \mu)^{\mathbf{2}}$$

Taking expectations:

$$E[\hat{\sigma}^2_{\mathrm{MLE}}] = \frac{1}{n}\sum_{i=1}^{n}E[(X_i - \mu)^2] = \frac{1}{n}(n\sigma^2) = \sigma^{\mathbf{2}}$$

The estimator is strictly unbiased.

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